Antipsychotic drugs for the acute treatment of patients with a first episode of schizophrenia: a systematic review with pairwise and network meta-analyses

Yikang Zhu, Marc Krause, Maximilian Huhn, Philipp Rothe, Johannes Schneider-Thoma, Anna Chaimani, Chunbo Li, John M Davis, Stefan Leucht

Summary

Background **The first episode of schizophrenia is a pivotal phase of this debilitating illness. Which drug to use remains controversial without a summary of all direct or indirect comparisons of drugs. We did a systematic review with pairwise and network meta-analyses of efficacy and tolerability.**

Methods **We searched MEDLINE, Embase, PsycINFO, Cochrane Library, PubMed, Biosis, and ClinicalTrials.gov for randomised controlled trials of antipsychotics for the acute treatment of first-episode schizophrenia, published up to Nov 17, 2016. Our primary outcome was overall change in symptoms. Secondary outcomes were change in positive and negative symptoms, categorical response to treatment, study dropout for any reason and for inefficacy of treatment, use of drugs to treat parkinsonian symptoms, weight gain, sedation, increase in prolactin release, overall functioning, and quality of life. We did the meta-analyses with a random-effects model to calculate standardised mean differences (SMDs) or odds ratios (ORs) with 95% CIs.**

Findings **We identified 19 relevant randomised controlled trials of 12 antipsychotic drugs that involved 2669 participants. 13 of the studies presented data on the primary outcome. For overall reduction of symptoms, amisulpride (SMD –0·37, 95% CI –0·61 to –0·14), olanzapine (–0·25, –0·39 to –0·12), ziprasidone (–0·25, –0·48 to –0·01), and risperidone (–0·14, –0·27 to –0·01) were significantly more efficacious than haloperidol, but the evidence was very low to moderate quality. Amisulpride was superior for reduction of symptoms to quetiapine (SMD –0·25, 95% CI –0·50 to –0·01). Olanzapine was superior to haloperidol and risperidone for reduction of negative symptoms. Several second-generation antipsychotics were superior to haloperidol in terms of all-cause discontinuation. Olanzapine was associated with at least one use of drugs to treat parkinsonian symptoms and quetiapine with less akathisia than haloperidol, aripiprazole, risperidone, and olanzapine, but, again, evidence was very low to low quality. Molindone was superior to risperidone, haloperidol, and olanzapine in terms of weight gain, and superior to risperidone in terms of increase in prolactin release.**

Interpreation **Haloperidol seems to be a suboptimum treatment option for acute treatment of first-episode schizophrenia, but we found little difference between second-generation antipsychotics. The evidence was generally of low quality and the numbers of patients for each drug were small. Thus, the choice of treatment should be guided primarily by side-effects.**

Funding **German Federal Ministry of Education and Research.**

Lancet Psychiatry 2017

Published Online July 20, 2017 http://dx.doi.org/10.1016/S2215-0366(17)30270-5

See Online/Comment http://dx.doi.org/10.1016/S2215-0366(17)30288-2

Department of Psychiatry and Psychotherapy, Technische Universität München, Klinikum rechts der Isar, Munich, Germany (Y Zhu MMed, M Krause MA, M Huhn MD, P Rothe MD, J Schneider-Thoma MD, Prof S Leucht MD); **Shanghai Key Laboratory of Psychotic Disorders, Shanghai Mental Health Centre, Shanghai Jiao Tong University School of Medicine, Shanghai, China** (Y Zhu, Prof C Li MD); **Paris Descartes University, Paris, France** (A Chaimani PhD); **INSERM, UMR1153 Epidemiology and Statistics, Sorbonne Paris Cité Research Center (CRESS), Paris, France** (A Chaimani); **Cochrane France, Paris, France** (A Chaimani); **Psychiatric Institute, University of Illinois at Chicago, Chicago, IL, USA** (Prof J M Davis MD); **and Maryland Psychiatric Research Center, Baltimore, MD, USA** (Prof J M Davis)

Correspondence to: Prof Stefan Leucht, Department of Psychiatry and Psychotherapy, Technische Universität München, Klinikum rechts der Isar, Ismaningerstrasse 22, 81675 Munich, Germany **stefan.leucht@tum.de**

Introduction

Schizophrenia is a severe mental disorder that affects more than 21 million people worldwide. WHO has ranked the disorder the eighth leading cause of disability among all illnesses worldwide in the age group 15–44 years.[1] According to the 2015 Global Burden of Disease Study, the total number of disability-adjusted life-years associated with schizophrenia has risen by more than 17% since 2005.[2]

The first episode of schizophrenia is particularly important for several reasons. Characteristics of patients differ from those of patients with chronic disease in various ways. First episodes generally occur in people aged 15–25 years and are associated with less pronounced negative symptoms than chronic schizophrenia.[3] Some cognitive domains, such as working memory and social cognition, are more preserved than in chronic disease.[4] Neuroimaging studies suggest that brain volume loss[5] and functional connectivity alterations[6] are less in the first episode than later in the disease course.

The duration of untreated psychosis is negatively associated with long-term outcomes.[7,8] People presenting with a first episode of schizophrenia usually respond very well to antipsychotic drugs.[9–12] By contrast, the effects of antipsychotics in chronic patients can be unsatisfactory, although a good response is seen again in some patients when relapse happens. Meta-analyses of treatment in chronic patients have shown moderate differences in efficacy for different antipsychotics compared with placebo (median standardised mean difference [SMD] 0·44, range 0·33–0·88)[13] and that only 23% of patients treated with antipsychotics versus 14% receiving placebo achieved at least 50% reduction in Positive and Negative Syndrome Scale score (PANSS) or the Brief Psychiatric Rating Scale (BPRS) from baseline or a Clinical Global Impression Rating of at least much

Research in context

Evidence before this study

The first episode of schizophrenia is a pivotal phase of this illness, but the choice of which drug to use is controversial. We searched MEDLINE and the Cochrane Library for analyses of drugs comparisons and found two conventional pairwise meta-analyses, one showing no difference between first-generation and second-generation antipsychotics, and one showing significant superiority of individual second-generation antipsychotics for specific factors. We found no systematic reviews that compared later-introduced antipsychotics with each other or that used a network meta-analytic approach. We searched MEDLINE, Embase, PsycINFO, Cochrane Library, PubMed, Biosis, and ClinicalTrials.gov for all randomised controlled trials published up to Nov 17, 2016, that compared antipsychotic drugs with each other or with placebo, from which we extracted data on first-episode patients.

Added value of this study

We used network meta-analysis to compare all antipsychotics with data available from randomised controlled trials on the acute treatment of first-episode schizophrenia in various domains of efficacy and tolerability. We included 19 randomised controlled trials involving 2669 participants, published between 1987 and 2015. Several second-generation antipsychotics were superior to haloperidol in terms of efficacy, acceptability, and some aspects of tolerability, whereas we found few and inconsistent, differences between individual second-generation antipsychotics. Tolerability overall was similar to that in patients receiving chronic treatment.

Implications of all the available evidence

Second-generation antipsychotics are the mainstay of treatment in developed countries, and we found little evidence on which to base drug choice in terms of efficacy in patients presenting with first-episode schizophrenia. Therefore, treatment decisions in this population should be guided by side-effects.

improved.[14] Therefore, optimum treatment of the first episode might improve the long-term prognosis.

In the past few decades, multiple randomised controlled trials of treatments for patients having a first episode of schizophrenia have been done. Two conventional pairwise meta-analyses have assessed whether second-generation antipsychotics are better than first-generation compounds to treat first episodes, but they came to different conclusions,[15,16] possibly because different approaches were used. The first compared first-generation and second-generation antipsychotics as groups and found no difference in efficacy between them, whereas the second compared each second-generation antipsychotics individually and found that several were superior to first-generation drugs. Many second-generation antipsychotics have lost patent protection and, therefore, increased cost no longer plays a major part in the choice of drug. Thus, comparisons of first-generation with second-generation antipsychotics have lessened, and it has even been suggested that they be merged into one class.[17] This strategy has been adopted by some international bodies, such as the European and American Colleges of Neuropsychopharmacology, which now use Neuroscience based Nomenclature[18] to classify antipsychotics by primary mechanism of action. Consequently, differences between individual antipsychotics have become more important than whether they are first or second generation. Because so many antipsychotics are available, however, conventional pairwise meta-analyses, which can compare only two drugs at a time, do not yield robust data on which to base recommendations. Network meta-analysis might be particularly helpful in this regard, because it allows analysis of all the evidence simultaneously. Moreover, by using indirect and direct evidence, gaps in the evidence matrix can be filled.[19] We used this method to assess which antipsychotics are most suitable to treat first-episode schizophrenia in various domains of efficacy and tolerability.

For the **study protocol** see www.crd.york.ac.uk/prospero

See Online for appendix

For the **Cochrane Schizophrenia Group** see http://schizophrenia.cochrane.org

Methods

Study design and participants

The study protocol is registered at PROSPERO, number CRD42015025111 (appendix pp 2–10), and followed the PRISMA extension for network meta-analysis.[20,21] We included individuals presenting with a first episode of schizophrenia or related disorders (eg, schizophreniform or schizoaffective disorders), using the definitions of first episode made by the study authors. We placed no restrictions on age, setting, sex, or ethnicity. To meet the assumptions of the analyses, we optimised homogeneity of studies within and across treatment comparisons by excluding those in treatment-resistant patients, in people with predominantly negative symptoms or concomitant medical or psychiatric illnesses (eg, studies in which all patients also had concomitant cannabis misuse), and in stable patients (ie, mainly relapse-prevention studies). We accepted studies in which less than 20% of participants had psychiatric disorders other than schizophrenia (eg, depression or mental retardation) or less than 20% of participants were not having a first episode. Following the rules of the Cochrane Schizophrenia Group, we included the trials irrespective of the diagnostic criteria used.

Antipsychotics

We considered the following antipsychotic drugs: amisulpride, aripiprazole, asenapine, benperidol, brexpiprazole, cariprazine, chlorpromazine, clozapine, flupenthixol, fluphenazine, fluspirilene, haloperidol,

iloperidone, levomepromazine, loxapine, lurasidone, molindone, olanzapine, paliperidone, quetiapine, penfluridol, perazine, perphenazine, pimozide, risperidone, sertindole, sulpiride, thioridazine, tiotixene, trifluoperazine, ziprasidone, zotepine, and zuclopenthixol (also known as clopenthixol). In fixed-dose studies, we followed the International Consensus Study of Antipsychotic dosing,[22] which recommends 25–30% lower doses for first-episode patients than for chronic patients. We included all flexible-dose studies because these allow the investigators to titrate the dose to that adequate for the individual patient. We excluded studies of depot formulations because these are mainly used for long-term prevention of relapse, which is not addressed in this review.

Outcomes

The primary outcome was overall change in symptoms of schizophrenia as measured by rating scales, such as the PANSS,[23] the BPRS,[24] or any other validated scale (eg, the Manchester Scale[25]). As not all studies used the same scale, we applied the following hierarchy: first, mean change in the PANSS total score from baseline to endpoint; if unavailable, mean change in BPRS score; if change in PANSS or BPRS were unavailable, the mean scores for either at the endpoint; then other scales. Secondary outcomes were response (as defined in the study; if available, we preferred 50% reduction in PANSS or BPRS and Clinical Global Impression of at least much improved to lower thresholds[26]), change in positive symptoms of schizophrenia, change in negative symptoms of schizophrenia, study dropout for any reason (all-cause discontinuation), dropout because of inefficacy of treatment, use of drugs to treat parkinsonian symptoms, akathisia, weight gain (we extracted data on mean weight gain and weight gain for at least 7%, although in this study we analyse only mean change), increased prolactin release (we extracted data on mean change and number of participants with substantial increases, but analyse only mean change here), sedation, overall functioning, and quality of life. Because this review is on acute treatment, we only included the studies that provided short-term data (≤13 weeks).

Search strategy and selection criteria

We searched MEDLINE, Embase, PsycINFO, Cochrane Library, PubMed, Biosis, and ClinicalTrials.gov for randomised controlled trials published up to Nov 17, 2016, that compared antipsychotic drugs with each other or with placebo in people with schizophrenia. We also searched the reference lists of previous reviews.[13,15,16] The search terms included those related to schizophrenia and schizophrenia-like disorders, randomisation, and all aforementioned drugs (appendix pp 11–22). As this review is part of a project that involves several subgroups of schizophrenia patients, we did not set the search terms to find only first-episode patients.

We included all randomised controlled trials, but excluded quasi-randomised studies (eg, allocation by day of the week). Because the number of randomised controlled trials of treatment for first-episode schizophrenia was small, we also included open-label studies. In crossover trials, we only included data up to the point of the first crossover to avoid carry-over effects.[27] Cluster-randomised trials were excluded. We excluded studies from mainland China to avoid a systematic bias because many did not use appropriate randomisation procedures or report the details of their methods.[28] Moreover, the China Food and Drug Administration has reported that many of the published reports are not reliable.[29]

Data extraction and risk of bias assessment

All abstracts identified in the searches were reviewed independently by two of YZ, MK, MH, and an independent analyst. Disagreements were resolved by discussion. If doubt remained, the full paper was obtained for further assessment. Once eligible papers had been selected, the full reports were obtained for all papers, which were again independently reviewed by at least two of the same assessment group. Disagreements about eligibility were discussed with SL and the original authors were contacted by email to ask for further information.

Two of YZ, PR, and SL and an independent analyst reviewed the main reports and supplementary materials, extracted the relevant data and entered the information into electronic forms, and assessed risk of bias in terms of sequence generation, allocation concealment, blinding, completeness of outcome data, selective reporting, and other biases, with the Cochrane risk of bias tool.[30] We also assessed a global risk of bias rating for each study based on the criteria applied in a network meta-analysis of antidepressants.[31]

Statistical analysis

First, we did random-effects conventional pairwise meta-analyses with RevMan (version 5.3). We calculated SMDs for continuous outcomes and odds ratios (ORs) for binary outcomes, with 95% CIs. In line with Cohen's guideline for magnitude of effect, we took SMDs of –0·2 to be small, –0·50 to be medium, and –0·8 to be large.[32] The heterogeneity in each pairwise comparison was assessed with the I^2 statistic (I^2 >50% taken to indicate substantial heterogeneity).

Second, we did a random-effects network meta-analysis in Stata (version 14.0), using the network package to estimate summary effect sizes that were again presented as SMDs or ORs with 95% CIs.[33,34] Network meta-analysis integrates direct evidence (direct comparisons of two or more interventions) and indirect evidence (drawn from two interventions that have separately been compared with one or more common comparators) for every treatment contrast. For instance,

if there are trials that directly compare risperidone with aripiprazole and trials that compare risperidone with quetiapine, but none compares aripiprazole with quetiapine, network meta-analysis enables indirect estimation of the effect of aripiprazole versus quetiapine from the other comparisons. Moreover, network meta-analysis can create a hierarchy of treatments based on the surface under the cumulative ranking curve (SUCRA).[35] This is a simple transformation of the mean rank and ranges from 0 to 1, with 1 indicating that a treatment is certain to be the best and 0 that a treatment is certain to be the worst.

Before we did the network meta-analysis, we attempted to assess the transitivity assumption.[36] This assumption implies that studies comparing different sets of interventions are sufficiently similar to provide valid indirect inferences, which we hoped to ensure by applying narrow inclusion criteria, making populations similar within and across treatment comparisons. We also considered whether the potential effect modifiers were distributed similarly across the available direct comparisons.

We assumed a common heterogeneity estimate for all treatment comparisons. We presented the between-study variance τ^2 for each outcome and assessed heterogeneity by referring to empirical distributions of heterogeneity values typically found in meta-analyses.[37,38] Statistical inconsistency (disagreements between direct and indirect evidence) was tested with three different approaches: the loop-specific approach that tests inconsistency in every closed loop of evidence;[39] the side-splitting method that tests for each comparison discrepancies between direct and indirect evidence obtained by the entire network;[40] and the design-by-treatment interaction model that tests inconsistency from all possible sources in the network jointly.[41]

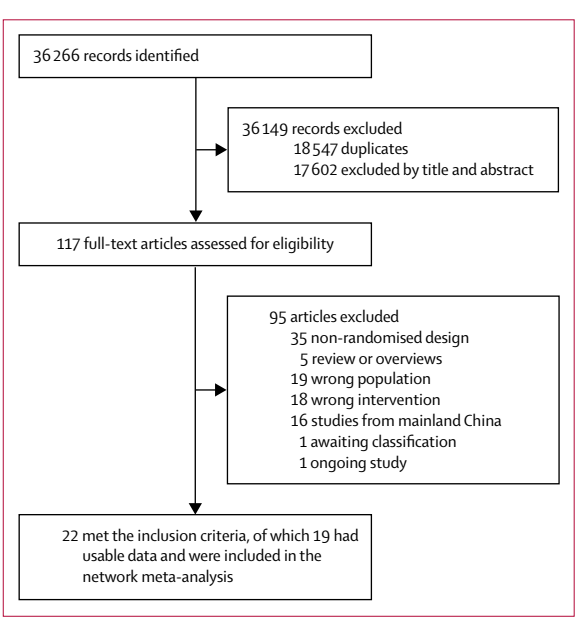

Figure 1: **Selection of studies**

For **CINeMA** see http://ec2-35-156-97-18.eu-central-1.compute.amazonaws.com:8004/ocpu/library/contribution/www/#welcome

We planned a priori two sensitivity analyses on the primary outcome, intended to assess the robustness of the estimates from the primary analysis. In one we excluded open-label randomised controlled trials, and in the other we excluded studies that did not use operationalised criteria, such as ICD-10 or DSM-IV. In response to peer review, we also did a post-hoc sensitivity analysis in which we included the study by Robinson and colleagues,[42] which was originally excluded because its duration (16 weeks) exceeded our predefined maximum of 13 weeks. Usable data in that study were available only for response and weight gain and, therefore, we analysed these instead of our primary outcome.

To explore reasons for heterogeneity, we planned meta-regression analyses (continuous moderators) and subgroup analyses (dichotomous moderators) of the primary outcome according to the following variables: antipsychotic naive, severity of illness at baseline, duration of untreated psychosis, and the ratio of male to female participants. In response to peer review, we also did a post-hoc subgroup analysis of overall symptom change and use of drugs to treat parkinsonian symptoms based on haloperidol dose (≥4 *vs* <4 mg per day). We did pairwise subgroup meta-analyses of different doses of haloperidol and risperidone (≥4 *vs* <4 mg per day) versus all other antipsychotics on these outcomes. Lower thresholds could not be assessed, because the lowest doses for haloperidol and risperidone, respectively, were 3·0 mg per day and 2·4 mg per day.

We investigated potential effects from small trials and publication bias for the primary outcome with a comparison-adjusted funnel plot.[43] We assumed that drugs introduced to the market later would be potentially favoured by small trials. We also planned to create standard funnel plots of single comparisons (eg, olanzapine *vs* haloperidol) if at least ten studies were available.

Finally, we assessed the quality of evidence for the primary outcome and the secondary outcome at least one use of drugs to treat parkinsonian symptoms with the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach, following the framework suggested by Salanti and colleagues.[19] We used the online tool, CINeMA, to assess study limitations in the following five domains for all relative effects and for treatment ranking estimated by network meta-analysis: study limitations, indirectness, inconsistency, imprecision, and publication bias.

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The corresponding author had full access to all data in the study and had final responsibility for the decision to submit for publication.

Results

We identified 36 266 potentially relevant citations, from which 117 were left after removal of duplicates and the titles and abstracts were reviewed (figure 1). After checking the full-text articles, we identified 19 randomised controlled trials involving 2669 participants that had us able data (table).[3,44–62] The reports were published between 1987 and 2015, and provided comparisons of 12 antipsychotic drugs that were included in the network meta-analysis. 11 studies were of haloperidol, 13 of risperidone, seven of olanzapine, four of quetiapine, and one each of ziprasidone, zuclopenthixol, molindone, flupenthixol, pimozide, aripiprazole, amisulpride, and sertindole. The mean sample size was 140 participants (range 8–498), the mean duration of illness was 1·5 years (SD 1·8), and the mean age of trial participants was 23·9 years (SD 9·7). Sex was indicated for 2598 patients, of whom 1710 (66%) were men. The median duration of trial was 8 weeks (range 4–13). The overall risk of bias findings are shown in the appendix (pp 23–25). Few details were reported about randomisation procedures and concealment of treatment allocation. 12 (63%) studies were double blind, three (16%) were single blind (those assessing outcomes were blinded), and four (21%) studies were open label. We judged five (26%) and two (11%) of the studies to have a high risk of bias in terms of attrition and selective reporting, respectively, and that only a few other studies had clear methodological problems, such as imbalance of groups at baseline. Nine (47%) studies were funded by pharmaceutical companies.

The mean antipsychotic dose was 12 mg per day in olanzapine equivalents.[22] The network of comparators for the primary outcome is shown in figure 2, and those for secondary outcomes are presented in the appendix (pp 26–37). The results of simple pairwise and network meta-analyses were generally consistent, except that, as expected, network meta-analysis produced more

	Study treatments (number of patients)	Trial duration (weeks)	Mean dose (DDD, MED, CMD, IC; mg per day)	Diagnosis	Definition of first episode	Study design	Risk of bias*	Characteristics of patients
Amr et al, 2013[45]	Haloperidol (n=33), quetiapine (n=40)	12	Haloperidol: 14·2 (17·8, 26·8, 19·2, 28·4); quetiapine: 705·8 (17·7, 35·3, 21·9, 19·1)	Schizophrenia (DSM-IV-TR)	First-episode schizophrenia	SB-RCT	5	Outpatients; 46 (63%) men, 27 (37%) women; mean age 31·1 years (SD 3·7); mean duration of illness 5·0 months (SD 1·9)
Brewer et al, 2007[46]	Haloperidol (n=4), risperidone (n=4)	8	Haloperidol: 2·0 (2·5, 3·8, 2·7, 4·0); risperidone: 2·0 (4·0, 7·4, 5·3, 6·7)	Schizophrenia (DSM-IV)	Antipsychotic-naive first-episode schizophrenia	DB-RCT	3	Inpatients and outpatients; 8 (100%) men; mean age 21·2 years (SD 3·0); age of onset of symptoms 16–30 years; mean baseline PANSS score: total 50·3 (SD 7·5), positive symptoms 29·7 (2·5), negative symptoms 20·5 (6·9)
Chaudhuri et al, 2000[47]	Haloperidol (n=15), risperidone (n=15)	4	Haloperidol: 15·0 (18·8, 28·3, 20·3, 30·0); risperidone: 4·0 (8·0, 14·8, 10·5, 13·3)	Acute and transient psychotic disorder (ICD-10)	Antipsychotic-naive first-episode schizophrenia	SB-RCT	2	Inpatients; 15 (50%) men, 15 (50%) women; age groups: 16–25 years (n=24), 26–35 years (n=5), 46–55 years (n=1); no history of psychiatric morbidity
Crespo-Facorro et al, 2006[48]	Haloperidol (n=56), olanzapine (n=55), risperidone (n=61)	6	Haloperidol: 5·4 (6·8, 10·2, 7·3, 10·8); olanzapine: 15·3 (15·3, 15·3, 15·3, 15·3); risperidone: 4·0 (8·0, 14·8, 10·5, 13·3)	Schizophreniform disorder, schizophrenia, schizoaffective disorder, brief reactive psychosis, schizotypal personality disorder, or psychosis not otherwise specified (DSM-IV)	First-episode or <6 weeks' antipsychotic treatment	OL-RCT	4	Inpatients and outpatients; 107 (62%) men, 65 (38%) women; mean age 27·3 years (SD 7·8); mean duration of illness 27·9 months (SD 36·7); moderate or worse baseline psychotic symptoms of moderate severity or greater assessed (≥1 of 5 items on SAPS)
Emsley et al, 1999[49]	Haloperidol (n=84), risperidone (n=99)	6	Haloperidol: 5·6 (7·0, 10·6, 7·6, 11·2); risperidone: 6·1 (12·2, 22·6, 16·1, 20·3)	Schizophreniform disorder or schizophrenia (DSM-III-R)	Antipsychotic-naive first-episode schizophrenia	DB-RCT	3	Inpatients and outpatients; 122 (67% men), 61 (33%) women; median age 26 years in risperidone group and 24 years in haloperidol group
Fagerlund et al, 2004[50]	Risperidone (n=15), zuclopenthixol (n=10)	13	Risperidone: 3·6 (7·2, 13·3, 9·5, 12·0); zuclopenthixol: 9·6 (3·2, NC, NC, 3·8)	Schizophrenia (ICD-10)	Antipsychotic-naive first-episode schizophrenia	OL-RCT	3	Inpatients; mean age 27·3 years (SD 5·9); median duration of untreated psychosis 14 months
Gafoor et al, 2010[51]	Quetiapine (n=38), risperidone (n=34)	12	Quetiapine: 375·0 (9·4, 18·8, 11·6, 10·1); risperidone: 2·72 (5·4, 10·1, 7·2, 9·1)	Schizophrenia spectrum disorders (ICD-10)	First-episode with <2 weeks' antipsychotic treatment	SB-RCT	4	Outpatients; 52 (72%) men, 20 (28%) women; mean 24·0 years (SD 4·9); baseline mean scores: PANSS total 70·4 (SD 22·0), CGI 4·94 (SD 0·92)
Gallhofer et al, 2007[52]	Sertindole (n=13), haloperidol (n=13)	12	Sertindole: 11·8 (7·4, 7·4, 10·9, 11·8); haloperidol: 5·8 (7·3, 10·9, 7·8, 11·6)	Schizophreniform disorder, acute schizophrenia, or schizophrenia (DSM-IV)	First-episode schizophrenia	DB-RCT	3	6 (25%) men, 18 (75%) women; mean age 28·1 years (SD 8·2); mean baseline PANSS total score 68·8 (SD 21·6); mean duration of illness 0·07 years (SD 0·22)

(Table continues on next page)

	Study treatments (number of patients)	Trial duration (weeks)	Mean dose (DDD, MED, CMD, IC; mg per day)	Diagnosis	Definition of first episode	Study design	Risk of bias*	Characteristics of patients
(Continued from previous page)								
Fleischhacker et al, 2012,[61] Kahn et al, 2008[62]	Haloperidol (n=103), amisulpride (n=104), olanzapine (n=105), quetiapine (n=104), ziprasidone (n=82)	12 (52†)	Haloperidol: 3·0 (3·8, 5·7, 4·1, 6·0); amisulpride: 450·8 (11·3, NC, 11·8, 13·1); olanzapine: 12·6 (12·6, 12·6, 12·6, 12·6); quetiapine: 498·6 (12·5, 24·9, 15·5, 13·5); ziprasidone: 107·2 (13·4, 20·1, 13·5, 13·4)	Schizophrenia, schizophreniform disorder, or schizoaffective disorder (DSM-IV)	First-episode schizophrenia	OL-RCT	5	Inpatients and outpatients; 298 (60%) men, 200 (40%) women; mean age 26 years (SD 5·6); mean baseline PANSS total score 88·5 (SD 20·6)
Lee et al, 2007[53]	Haloperidol (n=10), risperidone (n=10)	8	Haloperidol: 7·6 (9·5, 14·3, 10·3, 15·2); risperidone: 4·1 (8·2, 15·2, 10·8, 13·7)	Schizophrenia (DSM-IV)	Antipsychotic-naive schizophrenia	DB-RCT	2	Inpatients; 20 (100%) men; mean age 26·6 years (SD 8·8); mean baseline PANSS total 92·3 (SD 12·2)
Lieberman et al, 2003[54]	Olanzapine (n=131), haloperidol (n=132)	12	Olanzapine: 9·1 (9·1, 9·1, 9·1, 9·1); haloperidol: 4·4 (5·5, 8·3, 5·9, 8·8)	Schizophrenia, schizophreniform disorder, or schizoaffective disorder (DSM-IV)	First-episode schizophrenia	DB-RCT	3	Inpatients and outpatients; 215 (82%) men, 48 (18%) women; mean age 23·8 years (SD 4·8); age of onset of psychotic symptoms <35 years; score ≥4 on two or more PANSS psychosis items or ≥5 on one psychosis item and CGI severity score ≥4
McEvoy et al, 2007[55]	Olanzapine (n=133), quetiapine (n=134), risperidone n=133	12 (52†)	Olanzapine: 11·7 (11·7, 11·7, 11·7, 11·7); quetiapine: 506 (12·7, 25·3, 15·7, 13·7); risperidone: 2·4 (4·8, 8·9, 6·3, 8·0)	Schizophrenia, schizophreniform disorder, or schizoaffective disorder (DSM-IV)	First-episode schizophrenia, continuously ill for ≥1 month and ≤5 years	DB-RCT	5	Inpatients and outpatients; 292 (73%) men, 108 (27%) women; mean age 24·5 years (SD 5·8); mean duration of illness 12·9 months (SD 17·29); score ≥4 on one or more PANSS psychosis item and CGI severity score ≥4
Möller et al, 2008[56]	Risperidone (n=143), haloperidol (n=146)	8	Risperidone: 3·8 (7·6, 14·1, 10·0, 12·7); haloperidol: 3·7 (4·6, 7·0, 5·0, 7·4)	Schizophrenia (ICD-10)	First-episode schizophrenia	DB-RCT	6	Inpatients; 172 (60%) men, 117 (40%) women; mean age 30·1 years (SD 9·8); mean baseline PANSS total score 79·1 (SD 24·0)
Robinson et al, 2015[57]	Aripiprazole (n=106), risperidone (n=103)	12	Aripiprazole: 14·8 (9·9, 11·1, 10·5, 9·9); risperidone: 3·2 (6·4, 11·9, 8·4, 10·7)	Schizophrenia, schizophreniform, schizoaffective disorder, or psychotic disorder not otherwise specified (DSM-IV)	First-episode schizophrenia	DB-RCT	6	140 (71%) men, 58 (29%) women; mean age 22·1 years (SD 5·6); mean duration of psychotic symptoms 125·5 weeks (SD 208·8)
San et al, 2012[58]	Olanzapine (n=25), quetiapine (n=23), risperidone (n=25), ziprasidone (n=20), haloperidol (n=21)	12 (52†)	Olanzapine: 12·0 (12·0, 12·0, 12·0, 12·0); quetiapine: 572·0 (14·3, 28·6, 17·7, 15·4); risperidone: 3·7 (7·4, 13·7, 9·7, 12·3); ziprasidone: 81·0 (10·1, 15·2, 10·2, 10·1); haloperidol: 4·0 (5·0, 7·5, 5·4, 8·0)	Schizophrenia, schizophreniform disorder, schizoaffective disorder, psychotic disorder NOS, brief psychotic disorder (DSM-IV-TR)	Antipsychotic-naive first-episode schizophrenia	OL-RCT	5	Inpatients; 85 (75%) mean, 29 (25%) women; mean age 25·6 years (SD 8·0); mean baseline PANSS total score 91·0 (SD 20·0), mean duration of untreated psychosis 52·5 weeks (SD 170·0)
Sanger et al, 1999[3]	Haloperidol (n=24), olanzapine (n=59)	6	Haloperidol: 10·8 (13·5, 20·4, 14·6, 21·6); olanzapine: 11·6 (11·6, 11·6, 11·6, 11·6)	Schizophrenia, schizoaffective disorder, or schizophreniform disorder (DSM-III-R)	First-episode schizophrenia, current episode ≤5 years	DB-RCT	3	57 (69%) men, 26 (31%) women; mean age 28·5 years (SD 7·3); mean duration of illness 1·3 years and length of current episode 389·6 days (SD 422·8)
Scottish First Episode, 1987[44]	Flupenthixol (n=23), pimozide (n=23)	5	Flupenthixol: 20·0 (33·3, NC, NC, 40·0); pimozide: 18·8 (47·0, NC, NC, 47·0)	Schizophrenia	Antipsychotic-naive first-episode schizophrenia	DB-RCT	2	Inpatients; mean age 30·6 years (range 16–68); mean length of the first admission 11·8 weeks (SD 10·0)
Sikich et al, 2008[59]	Molindone (n=40), olanzapine (n=35), risperidone (n=41)	8	Molindone: 59·9 (12·0, NC, NC, 12·0); olanzapine: 11·4 (11·4, 11·4, 11·4, 11·4); risperidone: 2·8 (5·6, 10·4, 7·4, 9·3)	Schizophrenia, schizoaffective disorder, or schizophreniform disorder (DSM-IV [KID-SCID])	First-episode schizophrenia (early onset)	DB-RCT	7	Inpatients and outpatients; 75 (65%) men, 41 (35%) women; age range 8–19 years; at least moderate positive psychotic symptoms at enrolment
Svestka et al, 2003[60]	Olanzapine (n=21), risperidone (n=21)	6	Olanzapine: 18·0 (18·0, 18·0, 18·0, 18·0); risperidone: 4·9 (9·8, 18·1, 12·9, 16·3)	Schizophrenic and schizoform disorders (ICD-10)	First-episode schizophrenia	DB-RCT	2	Inpatients; 42 (100%) women; mean age 28·4 years; mean duration of episode 103·6 days (range 14·0–725·0)

DDD=defined daily doses method (olanzapine equivalent dose). MED=minimum effective dose method (olanzapine equivalent dose). CMD=classic mean dose method (olanzapine equivalent dose). IC=international consensus (olanzapine equivalent dose). TR=text revision. SB=single-blind. RCT=randomised controlled trial. DB=double-blind. PANSS=Positive and Negative Syndrome Scale. OL=open-label. SAPS=Scale for the Assessment of Positive Symptoms. R=revised. NC=not calculable by method. CGI=Clinical Global Impression scale. NOS=not otherwise specified. KID=SCID=Structured Clinical Interview for DMS-IV Childhood Diagnoses. *Number of low-risk judgments. †Long-term data also reported, but only short-term data assessed.

***Table*: Description of included studies**

significant findings. Therefore, the results of these conventional pairwise meta-analyses are always reported together. We used the SUCRA rankings (appendix pp 45–56) to order the results for these analyses.

For the primary outcome, mean score reduction in overall symptoms of schizophrenia, amisulpride, olanzapine, ziprasidone, and risperidone were significantly more efficacious than haloperidol, and amisulpride was significantly more efficacious than quetiapine (figure 3). In terms of all-cause discontinuation, aripiprazole, quetiapine, risperidone, and olanzapine were superior to haloperidol (figure 3). For dropout due to inefficacy, olanzapine and risperidone were superior to haloperidol (appendix p 41).

The network meta-analysis revealed no significant differences between treatments for change in positive symptoms, but for change in negative symptoms, olanzapine was significantly more efficacious than haloperidol and risperidone (figure 4). Categorical response to treatment did not differ between treatments (appendix p 40).

Olanzapine was associated with less frequent use of drugs to treat parkinsonian symptoms than haloperidol, zuclopenthixol, and risperidone, and quetiapine was associated with less use of drugs to treat parkinsonian symptoms than haloperidol and zuclopenthixol (figure 5). Molindone was associated with significantly less weight gain than olanzapine, haloperidol, and risperidone, and haloperidol was superior to olanzapine (figure 5).

Quetiapine was associated with less akathisia than haloperidol, aripiprazole, risperidone, and olanzapine, and olanzapine with less than haloperidol, aripiprazole, and risperidone (appendix p 42). Quetiapine was associated with less sedation than risperidone and aripiprazole (appendix p 43). Molindone, aripiprazole, olanzapine, and haloperidol were associated with lower increases in prolactin release than risperidone, as were molindone and olanzapine than haloperidol (appendix p 44). Overall functioning and quality of life outcomes did not differ between drugs in pairwise meta-analyses, but very few data were available. No estimate was produced by network meta-analyses.

When we excluded open-label studies from the sensitivity analysis for the primary outcome, olanzapine was superior to haloperidol and quetiapine in the network meta-analysis. When the 16-week study by Robinson and colleagues[42] was included in the sensitivity analysis of response to treatment and weight gain, the overall results for these outcomes changed little (appendix pp 73–77).

Most outcomes showed low heterogeneity (appendix pp 57–72). None of the methods we used suggested important inconsistency for the primary outcome, but the loop-specific and side-splitting methods revealed some inconsistency for the secondary outcomes (changes in positive symptoms, negative symptoms, and sedation). The design-by-treatment interaction model suggested that the transivity assumption might not be plausible for sedation and increase in prolactin release.

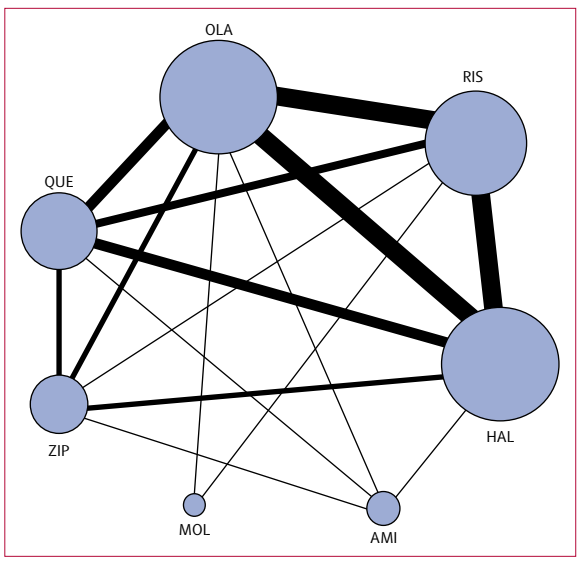

***Figure 2*: Network plot of eligible comparisons for overall change in symptoms**
The circles (nodes) represent the available treatments and the lines (edges) represent the available comparisons. Sizes of nodes and width of edges indicate weighting according to the number of studies involved for each treatment and comparison, respectively. OLA=olanzapine. RIS=risperidone. HAL=haloperidol. AMI=amisulpride. MOL=molindone. ZIP=ziprasidone. QUE=quetiapine.

The network meta-regression analyses for severity of illness at baseline, duration of untreated psychosis, and ratio of the sexes that had been planned a priori were not feasible because too few data were available. The subgroup analyses for antipsychotic-naive patients versus previously treated patients and for haloperidol dose did not show any clear differences (appendix pp 78–82), but interpretation was limited by small group size and differences in the antipsychotics used. The pairwise meta-analyses of overall change in symptoms and at least one use of drugs to treat parkinsonian symptoms by haloperidol and risperidone dose showed that higher haloperidol doses were associated with significantly more use of drugs to treat parkinsonian symptoms than lower doses ($p<0\cdot001$). Risperidone dose had no effect on this outcome. We found no differences for overall change in symptoms with either haloperidol or risperidone dose subgroup (appendix pp 83–87).

The comparison-adjusted funnel plot did not provide clear evidence that antipsychotics introduced to the market later were favoured by small studies compared with those introduced earlier (appendix pp 88–89). Funnel plots for individual comparisons were not meaningful because the maximum number of trials in a comparison was six.

The GRADE approach showed that the quality of the evidence for the outcomes overall change in symptoms and at least one use of drugs to treat parkinsonian symptoms was very low to moderate, depending on the individual comparison. The quality of the ranking was low for these two outcomes (appendix pp 90–95).

ARI	NMA 1·14 (0·45 to 2·87)	NMA 1·17 (0·43 to 3·17)	NMA 1·34 (0·76 to 2·37) PWA 1·34 (0·76 to 2·37)*	NMA 1·38 (0·66 to 2·89)	NMA 3·75 (0·58 to 24·32)	**NMA 2·53 (1·31 to 4·92)**
AMI	QUE	NMA 1·03 (0·36 to 2·97)	NMA 1·18 (0·57 to 2·47)	NMA 1·22 (0·57 to 2·61)	NMA 3·31 (0·51 to 21·33)	**NMA 2·23 (1·16 to 4·28) PWA 2·23 (1·16 to 4·28)***
NMA −0·12 (−0·35 to 0·12) PWA −0·23 (−0·50 to 0·04)*	OLA	MOL	NMA 1·15 (0·51 to 2·60) PWA 0·78 (0·32 to 1·92)*	NMA 1·18 (0·52 to 2·68) PWA 1·78 (0·71 to 4·46)*	NMA 3·20 (0·46 to 22·21)	NMA 2·16 (0·94 to 4·98)
NMA −0·09 (−0·53 to 0·36)	NMA −0·03 (−0·36 to 0·42) PWA 0·02 (−0·43 to 0·48)*	MOL	RIS	NMA 1·03 (0·64 to 1·65) PWA 1·35 (0·34 to 5·30)†	NMA 2·79 (0·47 to 16·55)	**NMA 1·88 (1·34 to 2·65) PWA 1·69 (1·17 to 2·43)‡**
NMA −0·13 (−0·40 to 0·15) PWA −0·10 (−0·39 to 0·19)*	NMA −0·01 (−0·24 to 0·22) PWA −0·03 (−0·46 to 0·40)†	NMA −0·04 (−0·49 to 0·40)	ZIP	OLA	NMA 2·72 (0·45 to 16·32)	**NMA 1·83 (1·23 to 2·74)** PWA 2·17 (0·94 to 5·02)§
NMA −0·23 (−0·48 to 0·02)	NMA −0·11 (−0·26 to 0·03) PWA −0·09 (−0·27 to 0·10)¶	NMA −0·15 (−0·53 to 0·24) PWA −0·15 (−0·58 to 0·29)*	NMA −0·10 (−0·35 to 0·14) PWA 0·05 (−0·54 to 0·64)*	RIS	SER	NMA 0·68 (0·12 to 3·87) PWA 0·68 (0·12 to 3·87)*
NMA −0·25 (−0·50 to −0·01) PWA −0·21 (−0·48 to 0·06)*	NMA −0·13 (−0·30 to 0·03) PWA −0·15 (−0·36 to 0·07)§	NMA −0·17 (−0·58 to 0·25)	NMA −0·13 (−0·37 to 0·12)	NMA −0·02 (−0·20 to 0·16) PWA −0·16 (−0·42 to 0·10)†	QUE	HAL
NMA −0·37 (−0·61 to −0·14) PWA −0·33 (−0·60 to −0·06)*	**NMA −0·25 (−0·39 to −0·12) PWA −0·29 (−0·50 to −0·09)¶**	NMA −0·29 (−0·69 to 0·11)	**NMA −0·25 (−0·48 to −0·01)** PWA −0·25 (−0·52 to 0·01)†	**NMA −0·14 (−0·27 to −0·01)** PWA −0·10 (−0·25 to 0·06)‡	NMA −0·12 (−0·29 to 0·05)	HAL

All-cause discontinuation (OR, 95% CI) — Treatment comparator
Treatment comparator — Overall change in symptoms (SMD, 95% CI)

Figure 3: **Overall change in symptoms and all-cause discontinuation**

Treatments are ranked by the surface under the cumulative ranking probabilities. Bold values are significant. Comparisons between treatments should be read from left to right, with the relevant estimate being in the common cell for the column-defining treatment and the row-defining treatment. For reduction in overall symptoms, SMDs <0 indicate that the treatment specified in the column is more efficacious than that in the row. For all-cause discontinuation, ORs >1 indicate that the treatment specified in the row is more efficacious than that in the column. To obtain SMDs for comparisons in the opposite direction, negative values should be converted into positive values, and vice versa. To obtain ORs for comparisons in the opposite direction, reciprocals should be taken. SMD=standardised mean difference. NMA=network meta-analysis. PWA=pairwise analysis. OR=odds ratio. ARI=aripiprazole. AMI=amisulpride. QUE=quetiapine. OLA=olanzapine. MOL=molindone. RIS=risperidone. ZIP=ziprasidone. SER=sertindole. HAL=haloperidol. *Includes one direct comparison study. †Includes two direct comparison studies. ‡Includes six direct comparison studies. §Includes three direct comparison studies. ¶Includes five direct comparison studies.

Discussion

We used pairwise and network meta-analyses to summarise the evidence from randomised trials for efficacy and tolerability of antipsychotic drugs in the short-term treatment of first-episode schizophrenia. We found some significant differences between drugs in terms of efficacy and premature all-cause discontinuation, dropout for inefficacy, or both, mainly showing a disadvantage for haloperidol, but tolerability results were generally similar to those seen in patients with chronic schizophrenia.[13]

The network meta-analysis suggested significant superiority for amisulpride, olanzapine, ziprasidone, and risperidone over haloperidol in the reduction of overall symptoms, and of olanzapine over haloperidol in reduction of negative symptoms. This superiority of several second-generation antipsychotics over haloperidol is similar to the pattern seen in a conventional meta-analysis by Zhang and colleagues.[16] By contrast, little difference was seen between second-generation antipsychotics, with the only significant differences being amisulpride superior to quetiapine for overall reduction of symptoms and olanzapine superior to risperidone for reduction of negative symptoms. The superiority of amisulpride over quetiapine was based on one open-label trial, and this result did not remain significant when open-label studies were excluded from the sensitivity analysis of the primary outcome. The only significant finding after this analysis was superiority of olanzapine over haloperidol and quetiapine. We conclude, therefore, that there is little evidence that treatment recommendations for second-generation antipsychotics, which are the most frequently used drugs in developed countries, can be based on differences in efficacy.

A possible explanation for the little difference between second-generation antipsychotics could be that patients having a first episode generally respond better to treatment than patients who have been taking antipsychotics long term, which might lead to ceiling effects. For example, in a meta-analysis of 167 randomised controlled trials involving 28 102 patients mainly with chronic disease, only 51% of drug-treated patients had minimal responses (defined as reductions of ≥20% in PANSS in BPRS or at least minimum improvement in Clinical Global Impression rating from baseline) and 23% had good responses (defined as reductions of ≥50% in PANSS or BPRS or a Clinical Global Impression rating of at least much improved from baseline).[14] By contrast, among first-episode patients, Robinson and colleagues[63] and Lieberman and colleagues[64] reported that 87% and 90% achieved remission by 1 year and 2 years, respectively. Such high remission rates might leave little leeway for differences between drugs.

In terms of all-cause discontinuation, most of the second-generation antipsychotics analysed—aripiprazole, quetiapine, risperidone, and olanzapine—were superior to haloperidol, and the ORs were high. Because all-cause discontinuation includes patients who drop out of studies due to inefficacy and side-effects, it is widely viewed as a measure of overall acceptability.[65] Our findings, therefore, suggest that second-generation antipsychotics are more acceptable than haloperidol for treatment of a first episode.

We feel that the results on side-effects in our network meta-analyses are compatible to some degree with those in patients with chronic disease,[13] although the doses used to treat first episodes were lower. In terms of use of drugs to treat parkinsonian symptoms as a broad measure of movement disorders, several second-generation antipsychotics were superior to the first-generation antipsychotics haloperidol and zuclopenthixol. Olanzapine was superior to risperidone, although this result was not unexpected because the latter drug is not free from extrapyramidal symptoms.[13] In terms of akathisia, quetiapine and olanzapine were superior not only to haloperidol but also to risperidone and aripiprazole, which indicates the associated low risk of movement disorders, especially with quetiapine, which was further superior to olanzapine. By contrast, risperidone and aripiprazole were clearly associated with akathisia in first-episode patients, although akathisia-like symptoms have been proposed to be an expression of the serotonergic effects of aripiprazole.[66]

Olanzapine was associated with more weight gain than all other second-generation antipsychotics in the network meta-analysis. Because of this feature, some guidelines do not recommend this drug for first-line use in people with a first episode of schizophrenia.[67,68] Risperidone, despite statistical inconsistency, was associated with most increase in prolactin release in our network meta-analysis, even more than haloperidol. No data were available for amisulpride and paliperidone, which have previously been associated with prolactin release.[13] This side-effect is important because it can be associated with amenorrhoea, galactorrhoea, hirsutism, gynaecomastia, erectile dysfunction, and osteoporosis. By contrast, the finding that quetiapine was less associated with sedation than aripiprazole and risperidone could be artifactual because a large network meta-analysis in patients with chronic schizophrenia has shown that quetiapine is the most sedating of these three drugs.[13] We found that the data for this outcome were significantly inconsistent, which suggests that the result is unreliable. Sedation is only assessed by open interviews in antipsychotic drug trials, and improved operationalisation might make the results more reliable.

Dose effects are extremely important.[69–71] We addressed this issue in several analyses that used doses as moderators of the outcomes overall change in symptoms and use of drugs to treat parkinsonian symptoms, for which dose effects might be the most relevant. We found no dose effects on the primary outcome, which is in line with some[13,72] but not other[69] previous meta-analyses in patients with chronic schizophrenia. For use of drugs to treat parkinsonian symptoms, however, we did find an association with haloperidol doses of 4 mg or more per

OLA	NMA 0·03 (–0·37 to 0·43) PWA –0·07 (–0·52 to 0·38)*	NMA 0·15 (–0·11 to 0·41) PWA 0·23 (–0·06 to 0·52)*	**NMA 0·20 (0·03 to 0·37)** PWA 0·13 (–0·08 to 0·34)†	**NMA 0·31 (0·13 to 0·48) PWA 0·34 (0·15 to 0·53)**†
OLA	MOL	NMA 0·12 (–0·33 to 0·58)	NMA 0·17 (–0·22 to 0·57) PWA 0·09 (–0·34 to 0·53)*	NMA 0·28 (–0·14 to 0·69)
NMA 0·00 (–0·50 to 0·51) PWA –0·02 (–0·47 to 0·44)*	MOL	QUE	NMA 0·05 (–0·20 to 0·30) PWA –0·14 (–0·44 to 0·15)*	NMA 0·16 (–0·13 to 0·44) **PWA 0·63 (0·16 to 1·11)***
NMA –0·04 (–0·28 to 0·21) PWA –0·01 (–0·22 to 0·19)†	NMA –0·04 (–0·54 to 0·46) PWA –0·06 (–0·49 to 0·38)*	RIS	RIS	NMA 0·11 (–0·06 to 0·27) PWA 0·03 (–0·13 to 0·19)‡
NMA –0·13 (–0·49 to 0·24) **PWA –0·36 (–0·65 to –0·06)***	NMA –0·13 (–0·73 to 0·47)	NMA –0·09 (–0·45 to 0·27) **PWA –0·33 (–0·62 to –0·03)***	QUE	HAL
NMA –0·24 (–0·49 to 0·00) PWA –0·16 (–0·40 to 0·08)†	NMA –0·25 (–0·78 to 0·28)	NMA –0·21 (–0·44 to 0·03) PWA –0·12 (–0·32 to 0·08)‡	NMA –0·12 (–0·50 to 0·27) **PWA –0·65 (–1·12 to –0·17)***	HAL

Change in negative symptoms (SMD, 95% CI) — Treatment comparator
Treatment comparator — Change in positive symptoms (SMD, 95% CI)

Figure 4: **Change in positive symptoms and negative symptoms**
Treatments are ranked by the surface under the cumulative ranking probabilities. Bold values are significant. Comparisons between treatments should be read from left to right, with the relevant estimate being in the common cell for the column-defining treatment and the row-defining treatment. For negative symptoms, SMDs >0 indicate that the treatment specified in the row is more efficacious than that in the column. For positive symptoms, SMDs <0 indicate that the treatment specified in the column is more efficacious than that in the row. To obtain SMDs for comparisons in the opposite direction, negative values should be converted into positive values, and vice versa. SMD=standardised mean difference. NMA=network meta-analysis. PWA=pairwise analysis. OLA=olanzapine. MOL=molindone. QUE=quetiapine. RIS=risperidone. HAL=haloperidol. *Includes one direct comparison study. †Includes three direct comparison studies. ‡Includes four direct comparison studies.

MOL	NMA 0·75 (−0·34 to 1·84)	NMA 0·83 (−0·21 to 1·86)	**NMA 0·93 (0·15 to 1·70)**	**NMA 1·00 (0·04 to 1·96) PWA 0·93 (0·47 to 1·39)***	**NMA 1·63 (0·83 to 2·44) PWA 1·77 (1·23 to 2·31)***
OLA	QUE	NMA 0·07 (−0·96 to 1·10)	NMA 0·17 (−0·59 to 0·94) PWA 0·19 (−0·26 to 0·63)*	NMA 0·24 (−0·97 to 1·46)	NMA 0·88 (−0·22 to 1·97)
NMA 1·01 (0·10 to 10·21)	QUE	ARI	NMA 0·10 (−0·59 to 0·79) PWA 0·10 (−0·18 to 0·38)*	NMA 0·17 (−1·00 to 1·33)	NMA 0·80 (−0·24 to 1·84)
NMA 0·31 (0·04 to 2·75)	NMA 0·31 (0·02 to 5·58)	ARI	RIS	NMA 0·07 (−0·87 to 1·01)	NMA 0·70 (−0·08 to 1·48)
NMA 0·24 (0·07 to 0·78) PWA 0·13 (0·05 to 0·36)†	NMA 0·24 (0·03 to 2·24)	NMA 0·77 (0·12 to 4·75) PWA 0·77 (0·43 to 1·38)*	RIS	HAL	**NMA 0·63 (0·11 to 1·16) PWA 0·64 (0·13 to 1·15)†**
NMA 0·10 (0·03 to 0·29) PWA 0·10 (0·02 to 0·38)‡	**NMA 0·10 (0·01 to 0·75) PWA 0·10 (0·03 to 0·29)***	NMA 0·31 (0·04 to 2·38)	NMA 0·41 (0·16 to 1·01) **PWA 0·32 (0·11 to 0·92)§**	HAL	OLA
NMA 0·02 (0·00 to 0·37)	**NMA 0·02 (0·00 to 0·66)**	NMA 0·07 (0·00 to 1·65)	NMA 0·09 (0·01 to 1·21) **PWA 0·09 (0·01 to 0·62)***	NMA 0·22 (0·01 to 3·47)	ZUC

Weight gain (SMD, 95% CI) — Treatment comparator
Treatment comparator — Use of antiparkinson drugs (OR, 95% CI)

Figure 5: **Weight gain and use of drugs to treat parkinsonian symptoms**
Treatments are ranked by the surface under the cumulative ranking probabilities. Bold values are significant. Comparisons between treatments should be read from left to right, with the relevant estimate being in the common cell for the column-defining treatment and the row-defining treatment. For use of drugs to treat parkinsonian symptoms, ORs <1 indicate that the treatment specified in the column is more efficacious than that in the row. For weight gain, SMDs >0 indicate that the treatment specified in the row is more efficacious than that in the column. To obtain SMDs for comparisons in the opposite direction, negative values should be converted into positive values, and vice versa. To obtain ORs for comparisons in the opposite direction, reciprocals should be taken. SMD=standardised mean difference. NMA=network meta-analysis. PWA=pairwise analysis. OR=odds ratio. MOL=molindone. OLA=olanzapine. QUE=quetiapine. ARI=aripiprazole. RIS=risperidone. HAL=haloperidol. ZUC=zuclopenthixol. *Includes one direct comparison study. †Includes two direct comparison studies. ‡Includes three direct comparison studies. §Includes four direct comparison studies.

day, but not with lower doses. As haloperidol carries a particularly high risk of movement disorders, however, the results in our network meta-analysis are not generalisable to other first-generation antipsychotics. Moreover, even in patients with chronic disease, the dose–response curve of haloperidol seems to start to flatten at about 4 mg per day[71] and in patients having a first episode, doses as low as 2 mg per day might be sufficient.[70] All the included studies assessed doses higher than 2 mg per day, which must be taken into account when interpreting the results. Dose-finding studies in first-episode patients might be useful to identify optimum antipsychotic doses.

This network meta-analysis had some strengths and limitations. Strengths were that the methods were stringent and followed the PRISMA guidelines, including publishing a protocol and the comprehensiveness of the outcome measures. Network meta-analysis has an advantage over conventional pairwise meta-analysis because it can synthesise direct and indirect evidence, which increases the precision of the estimates. However, although we included 19 trials with 2669 participants, multiple trials were available only for haloperidol, risperidone, olanzapine, and quetiapine. These drugs also had the most significant differences, whereas the results for all other drugs were based on data from single trials, meaning that many results were derived from indirect evidence. Moreover, the only first-generation antipsychotic for which several trials were available was haloperidol, whereas for molindone and pimozide only one randomised controlled trial was available for each. For the second-generation antipsychotics brexpiprazole, cariprazine, iloperidone, lurasidone, and paliperidone, which were the latest to enter the market, none had been assessed in a randomised controlled trial. In the network meta-analyses with few studies, the transitivity assumption is difficult to assess. We found no statistical inconsistency for the primary outcome, but we did see inconsistency in some secondary outcomes. Owing to the statistical power of the subgroup analyses in terms of duration of untreated psychosis,[8] antipsychotic dose, and antipsychotic-naive patients, the usefulness of the results was limited, but the effect of haloperidol dose on the use of drugs to treat parkinsonian symptoms was significant. Another limitation is that we only included short-term results of up to 13 weeks. A few long-term studies are available, but combining short-term and long-term studies in network meta-analysis would be difficult due to heterogeneity. Quality of life and functioning are becoming increasingly important in assessments of antipsychotic drugs, but very few data were available. These outcomes should be included in future studies. Finally, the GRADE approach used in the network meta-analysis[19] suggested only very low to moderate quality of evidence for overall symptoms and use of drugs to treat parkinsonian symptoms, which restricts the confidence that can be placed in the findings.

We found that most second-generation antipsychotics are superior to haloperidol in the domains of efficacy and acceptability, and in certain aspects of tolerability. Importantly, though, there were few significant differences between second-generation antipsychotics, which are the mainstay of treatment in many developed countries. Until findings are available from better-quality studies to elucidate differences in efficacy, treatment decisions for first-episode schizophrenia should be guided mainly by side-effects, for which the general patterns were similar to those found in chronic patients.

Contributors

YZ and SL designed the study. YZ, MK, and MH assessed the studies. YZ, SL, and PR extracted the data. YZ, MK, JS-T, and AC analysed the data. YZ and MK drew the tables and figures. YZ, CL, JMD, and SL interpreted data. YZ and SL wrote the first draft of the manuscript and all authors contributed to and have approved the final version.

Declaration of interests
SL has received honoraria for consulting from LB Pharma, Lundbeck, Otsuka, Roche, and TEVA, for lectures from AOP Orphan, ICON, Janssen, Lilly, Lundbeck, Otsuka, Pfizer, Roche, and Servier, and for a publication from Roche. MH has received speaker's honoraria from Janssen and Lundbeck. The other authors declare no competing interests.

Acknowledgments
This work was supported by a grant from the German Federal Ministry of Education and Research (Bundesministerium für Bildung und Forschung, BMBF, Grant number: FKZ 01KG1508). YZ was supported by Shanghai Sailing Program (16YF1416000), the Shanghai Jiao Tong University K C Wong Medical Fellowship Fund, and the visiting programme of the Shanghai Municipal Education Commission. YZ did this project as her doctoral thesis at Department of Psychiatry and Psychotherapy, Technische Universität München, Klinikum rechts der Isar, Munich, Germany, and this publication will be part of her dissertation. We thank Samantha Roberts for help with the literature search, Myrto T Samara for help in study selection, and Susanne Bächer for help in data extraction, and all authors of the included studies, particularly those who sent us additional information about their trials. The German Ministry of Education and Research paid a fee to Samantha Roberts.

References

1 WHO, The WHO world health report: new understanding, new hope. Geneva: World Health Organization, 2001.
2 GBD 2015 DALYs and HALE Collaborators. Global, regional, and national disability-adjusted life-years (DALYs) for 315 diseases and injuries and healthy life expectancy (HALE), 1990–2015: a systematic analysis for the Global Burden of Disease Study 2015. *Lancet* 2016; **388:** 1603–58.
3 Sanger TM, Lieberman JA, Tohen M, Grundy S, Beasley C Jr, Tollefson GD. Olanzapine versus haloperidol treatment in first-episode psychosis. *Am J Psychiatry* 1999; **156:** 79–87.
4 McCleery A, Ventura J, Kern RS, et al. Cognitive functioning in first-episode schizophrenia: MATRICS Consensus Cognitive Battery (MCCB) Profile of Impairment. *Schizophr Res* 2014; **157:** 33–39.
5 Torres US, Duran FL, Schaufelberger MS, et al. Patterns of regional gray matter loss at different stages of schizophrenia: a multisite, cross-sectional VBM study in first-episode and chronic illness. *Neuroimage Clin* 2016; **12:** 1–15.
6 Li T, Wang Q, Zhang J, et al. Brain-wide analysis of functional connectivity in first-episode and chronic stages of schizophrenia. *Schizophr Bull* 2017; **43:** 436–48.
7 Perkins D, Lieberman J, Gu H, et al. Predictors of antipsychotic treatment response in patients with first-episode schizophrenia, schizoaffective and schizophreniform disorders. *Br J Psychiatry* 2004; **185:** 18–24.
8 Perkins DO, Gu H, Boteva K, Lieberman JA. Relationship between duration of untreated psychosis and outcome in first-episode schizophrenia: a critical review and meta-analysis. *Am J Psychiatry* 2005; **162:** 1785–804.
9 Lieberman J, Jody D, Geisler S, et al. Time course and biologic correlates of treatment response in first-episode schizophrenia. *Arch Gen Psychiatry* 1993; **50:** 369–76.
10 Jager M, Riedel M, Messer T, et al. Psychopathological characteristics and treatment response of first episode compared with multiple episode schizophrenic disorders. *Eur Arch Psychiatry Clin Neurosci* 2007; **257:** 47–53.
11 Schennach R, Riedel M, Musil R, Moller HJ. Treatment response in first-episode schizophrenia. *Clin Psychopharmacol Neurosci* 2012; **10:** 78–87.
12 Zhu YK, Li CB, Huhn M, et al. How well do patients with a first episode of schizophrenia respond to antipsychotics: a systematic review and meta-analysis. *Eur Neuropsychopharamcol* 2017; published online June 29. DOI:10.1016/j.euroneuro.2017.06.011.
13 Leucht S, Cipriani A, Spineli L, et al. Comparative efficacy and tolerability of 15 antipsychotic drugs in schizophrenia: a multiple-treatments meta-analysis. *Lancet* 2013; **382:** 951–62.
14 Leucht S, Leucht C, Huhn M, et al. Sixty years of placebo-controlled antipsychotic drug trials in acute schizophrenia: systematic review, Bayesian meta-analysis, and meta-regression of efficacy predictors. *Am J Psychiatry* 2017; published online May 25. DOI:10.1176/appi.ajp.2017.16121358.
15 Crossley NA, Constante M, McGuire P, Power P. Efficacy of atypical v. typical antipsychotics in the treatment of early psychosis: meta-analysis. *Br J Psychiatry* 2010; **196:** 434–39.
16 Zhang JP, Gallego JA, Robinson DG, Malhotra AK, Kane JM, Correll CU. Efficacy and safety of individual second-generation vs. first-generation antipsychotics in first-episode psychosis: a systematic review and meta-analysis. *Int J Neuropsychopharmacol* 2013; **16:** 1205–18.
17 Tyrer P, Kendall T. The spurious advance of antipsychotic drug therapy. *Lancet* 2009; **373:** 4–5.
18 Zohar J, Stahl S, Moller HJ, et al. A review of the current nomenclature for psychotropic agents and an introduction to the Neuroscience-based Nomenclature. *Eur Neuropsychopharmacol* 2015; **25:** 2318–25.
19 Salanti G, Del Giovane C, Chaimani A, Caldwell DM, Higgins JP. Evaluating the quality of evidence from a network meta-analysis. *PLoS One* 2014; **9:** e99682.
20 Cornell JE. The PRISMA extension for network meta-analysis: bringing clarity and guidance to the reporting of systematic reviews incorporating network meta-analyses. *Ann Intern Med* 2015; **162:** 797–98.
21 Hutton B, Catala-Lopez F, Moher D. The PRISMA statement extension for systematic reviews incorporating network meta-analysis: PRISMA-NMA. *Med Clin (Barc)* 2016; **147:** 262–66 (in Spanish).
22 Gardner DM, Murphy AL, O'Donnell H, Centorrino F, Baldessarini RJ. International consensus study of antipsychotic dosing. *Am J Psychiatry* 2010; **167:** 686–93.
23 Kay SR, Fiszbein A, Opler LA. The positive and negative syndrome scale (PANSS) for schizophrenia. *Schizophr Bull* 1987; **13:** 261–76.
24 Overall JE, Gorham DR. The brief psychiatric rating scale. *Psychol Rep* 1962; **10:** 799–812.
25 Hyde CE. The Manchester scale. A standardised psychiatric assessment for rating chronic psychotic patients. *Br J Psychiatry Suppl* 1989; **7:** 45–48.
26 Guy W. ECDEU Assessment Manual for Psychopharmacology. Rockville, MD: US Department of Health, Education, and Welfare Public Health Service Alcohol, Drug Abuse, and Mental Health Administration, 1976.
27 Elbourne DR, Altman DG, Higgins JP, Curtin F, Worthington HV, Vail A. Meta-analyses involving cross-over trials: methodological issues. *Int J Epidemiol* 2002; **31:** 140–49.
28 Woodhead M. 80% of China's clinical trial data are fraudulent, investigation finds. *BMJ* 2016; **355:** i5396.
29 Center for Food and Drug Inspection of CFDA. Drug clinical trial data verification situation report [in Chinese]. www.cfdi.org.cn/resource/news/7713.html (accessed July 12, 2017) .
30 Higgins JP, Green S. Cochrane handbook for systematic reviews of interventions, version 5.1.0. Chichester: John Wiley & Sons, 2011.
31 Furukawa TA, Salanti G, Atkinson LZ, et al. Comparative efficacy and acceptability of first-generation and second-generation antidepressants in the acute treatment of major depression: protocol for a network meta-analysis. *BMJ Open* 2016; **6:** e010919.
32 Cohen J. Statistical power analysis for the behavioral sciences, 2nd edn. Mahwah, NJ: Lawrence Earlbaum Associates, 1988.
33 White IR. Network meta-analysis. *Stata J* 2015; **15:** 951–85.
34 Chaimani A, Salanti G. Visualizing assumptions and results in network meta-analysis: the network graphs package. *Stata J* 2015; **15:** 905–50.
35 Salanti G, Ades AE, Ioannidis JP. Graphical methods and numerical summaries for presenting results from multiple-treatment meta-analysis: an overview and tutorial. *J Clin Epidemiol* 2011; **64:** 163–71.
36 Salanti G. Indirect and mixed-treatment comparison, network, or multiple-treatments meta-analysis: many names, many benefits, many concerns for the next generation evidence synthesis tool. *Res Synth Methods* 2012; **3:** 80–97.
37 Rhodes KM, Turner RM, Higgins JP. Predictive distributions were developed for the extent of heterogeneity in meta-analyses of continuous outcome data. *J Clin Epidemiol* 2015; **68:** 52–60.
38 Turner RM, Davey J, Clarke MJ, Thompson SG, Higgins JP. Predicting the extent of heterogeneity in meta-analysis, using empirical data from the Cochrane Database of Systematic Reviews. *Int J Epidemiol* 2012; **41:** 818–27.

39 Bucher HC, Guyatt GH, Griffith LE, Walter SD. The results of direct and indirect treatment comparisons in meta-analysis of randomized controlled trials. *J Clin Epidemiol* 1997; **50:** 683–91.

40 Dias S, Welton NJ, Caldwell DM, Ades AE. Checking consistency in mixed treatment comparison meta-analysis. *Stat Med* 2010; **29:** 932–44.

41 Higgins JPT, Jackson D, Barrett JK, Lu G, Ades AE, White IR. Consistency and inconsistency in network meta-analysis: concepts and models for multi-arm studies. *Res Synth Methods* 2012; **3:** 98–110.

42 Robinson DG, Woerner MG, Napolitano B, et al. Randomized comparison of olanzapine versus risperidone for the treatment of first-episode schizophrenia: 4-month outcomes. *Am J Psychiatry* 2006; **163:** 2096–102.

43 Chaimani A, Higgins JP, Mavridis D, Spyridonos P, Salanti G. Graphical tools for network meta-analysis in STATA. *PLoS One* 2013; **8:** e76654.

44 The Scottish Schizophrenia Research Group. The Scottish First Episode Schizophrenia Study. II. Treatment: pimozide versus flupenthixol. *Br J Psychiatry* 1987; **150:** 334–38.

45 Amr M, Lakhan SE, Sanhan S, et al. Efficacy and tolerability of quetiapine versus haloperidol in first-episode schizophrenia: a randomized clinical trial. *Int Arch Med* 2013; **6:** 47.

46 Brewer WJ, Yucel M, Harrison BJ, et al. Increased prefrontal cerebral blood flow in first-episode schizophrenia following treatment: longitudinal positron emission tomography study. *Aust NZ J Psychiatry* 2007; **41:** 129–35.

47 Chaudhuri BP, Bhagabati D, Medhi D. Risperidone versus haloperidol in acute and transient psychotic disorder. *Indian J Psychiatry* 2000; **42:** 280–90.

48 Crespo-Facorro B, Perez-Iglesias R, Ramirez-Bonilla M, Martinez-Garcia O, Llorca J, Luis Vazquez-Barquero J. A practical clinical trial comparing haloperidol, risperidone, and olanzapine for the acute treatment of first-episode nonaffective psychosis. *J Clin Psychiatry* 2006; **67:** 1511–21.

49 Emsley RA. Risperidone in the treatment of first-episode psychotic patients: a double-blind multicenter study. Risperidone Working Group. *Schizophr Bull* 1999; **25:** 721–29.

50 Fagerlund B, Mackeprang T, Gade A, Glenthoj BY. Effects of low-dose risperidone and low-dose zuclopenthixol on cognitive functions in first-episode drug-naive schizophrenic patients. *CNS Spectr* 2004; **9:** 364–74.

51 Gafoor R, Landau S, Craig TK, Elanjithara T, Power P, McGuire P. Esquire trial: efficacy and adverse effects of quetiapine versus risperidone in first-episode schizophrenia. *J Clin Psychopharmacol* 2010; **30:** 600–06.

52 Gallhofer B, Jaanson P, Mittoux A, Tanghoj P, Lis S, Krieger S. Course of recovery of cognitive impairment in patients with schizophrenia: a randomised double-blind study comparing sertindole and haloperidol. *Pharmacopsychiatry* 2007; **40:** 275–86.

53 Lee SM, Chou YH, Li MH, Wan FJ, Yen MH. Effects of haloperidol and risperidone on cerebrohemodynamics in drug-naive schizophrenic patients. *J Psychiatr Res* 2008; **42:** 328–35.

54 Lieberman JA, Tollefson G, Tohen M, et al. Comparative efficacy and safety of atypical and conventional antipsychotic drugs in first-episode psychosis: a randomized, double-blind trial of olanzapine versus haloperidol. *Am J Psychiatry* 2003; **160:** 1396–404.

55 McEvoy JP, Lieberman JA, Perkins DO, et al. Efficacy and tolerability of olanzapine, quetiapine, and risperidone in the treatment of early psychosis: a randomized, double-blind 52-week comparison. *Am J Psychiatry* 2007; **164:** 1050–60.

56 Möller HJ, Riedel M, Jäger M, et al. Short-term treatment with risperidone or haloperidol in first-episode schizophrenia: 8-week results of a randomized controlled trial within the German Research Network on Schizophrenia. *Int J Neuropsychopharmacol* 2008; **11:** 985–97.

57 Robinson DG, Gallego JA, John M, et al. A randomized comparison of aripiprazole and risperidone for the acute treatment of first-episode schizophrenia and related disorders: 3-month outcomes. *Schizophr Bull* 2015; **41:** 1227–36.

58 San L, Arranz B, Perez V, et al. One-year, randomized, open trial comparing olanzapine, quetiapine, risperidone and ziprasidone effectiveness in antipsychotic-naive patients with a first-episode psychosis. *Psychiatry Res* 2012; **200:** 693–701.

59 Sikich L, Frazier JA, McClellan J, et al. Double-blind comparison of first- and second-generation antipsychotics in early-onset schizophrenia and schizo-affective disorder: findings from the treatment of early-onset schizophrenia spectrum disorders (TEOSS) study. *Am J Psychiatry* 2008; **165:** 1420–31.

60 Svestka J, Synek O, Zourkova A. Olanzapine versus risperidone in first-episode schizophrenic and schizoform disorders: a double-blind comparison. *Eur Neuropsychopharmacol* 2003; **13** (suppl 4)**:** S291.

61 Fleischhacker WW, Derks E, Kahn RS. Interpreting treatment trials in schizophrenia patients: lessons learned from EUFEST. *Schizophr Res* 2012; **138:** 39–40.

62 Kahn RS, Fleischhacker WW, Boter H, et al. Effectiveness of antipsychotic drugs in first-episode schizophrenia and schizophreniform disorder: an open randomised clinical trial. *Lancet* 2008; **371:** 1085–97.

63 Robinson DG, Woerner MG, Alvir JM, et al. Predictors of treatment response from a first episode of schizophrenia or schizoaffective disorder. *Am J Psychiatry* 1999; **156:** 544–49.

64 Lieberman JA, Phillips M, Gu H, et al. Atypical and conventional antipsychotic drugs in treatment-naive first-episode schizophrenia: a 52-week randomized trial of clozapine *vs* chlorpromazine. *Neuropsychopharmacology* 2003; **28:** 995–1003.

65 Cipriani A, Furukawa TA, Salanti G, et al. Comparative efficacy and acceptability of 12 new-generation antidepressants: a multiple-treatments meta-analysis. *Lancet* 2009; **373:** 746–58.

66 Thomas JE, Caballero J, Harrington CA. The incidence of akathisia in the treatment of schizophrenia with aripiprazole, asenapine and lurasidone: a meta-analysis. *Curr Neuropharmacol* 2015; **13:** 681–91.

67 Buchanan RW, Kreyenbuhl J, Kelly DL, et al. The 2009 schizophrenia PORT psychopharmacological treatment recommendations and summary statements. *Schizophr Bull* 2010; **36:** 71–93.

68 De Hert M, Yu W, Detraux J, Sweers K, van Winkel R, Correll CU. Body weight and metabolic adverse effects of asenapine, iloperidone, lurasidone and paliperidone in the treatment of schizophrenia and bipolar disorder: a systematic review and exploratory meta-analysis. *CNS Drugs* 2012; **26:** 733–59.

69 Geddes J, Freemantle N, Harrison P, Bebbington P. Atypical antipsychotics in the treatment of schizophrenia: systematic overview and meta-regression analysis. *BMJ* 2000; **321:** 1371–76.

70 Oosthuizen P, Emsley R, Jadri Turner H, Keyter N. A randomized, controlled comparison of the efficacy and tolerability of low and high doses of haloperidol in the treatment of first-episode psychosis. *Int J Neuropsychopharmacol* 2004; **7:** 125–31.

71 Zimbroff DL, Kane JM, Tamminga CA, et al. Controlled, dose-response study of sertindole and haloperidol in the treatment of schizophrenia. Sertindole Study Group. *Am J Psychiatry* 1997; **154:** 782–91.

72 Davis JM, Chen N, Glick ID. A meta-analysis of the efficacy of second-generation antipsychotics. *Arch Gen Psychiatry* 2003; **60:** 553–64.